

Separating Event Intensity from Fixed Funding Carry: Evidence from Perpetual Futures and Credit Markets

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Abstract

Perpetual futures combine a state-dependent anchoring premium with, on some venues, an exchange-specified fixed funding component. We ask whether the anchoring mechanism can be measured separately from fixed carry and whether its intensity has a coherent credit-market analogue. Using 804,000 hourly funding observations, 921 bond histories, 51 sovereign CDS series, and cross-domain loss data from seventeen providers, we obtain four results. Perpetual bases mean-revert in all 41 pre-specified dYdX markets, with a median half-life of 3.5 hours, and daily intensity estimates correlate across venues at 0.70. After harmonizing all venues to complete eight-hour intervals, signed mean funding is 0.83 basis points on dYdX, 1.02 on Binance, and 1.14 on Bybit across nine common markets. Mean drift therefore does not by itself identify the fixed input. The sharper diagnostic is distributional: centralized-exchange observations exhibit a pronounced mass around the one-basis-point formula point, whereas the premium-only venue does not. Spread-implied intensities are broadly ordered by rating and monotone in tenor and place the three subsequent sovereign default or restructuring events among the highest early-risk observations. Finally, a benchmark-free bond estimator preserves broad risk ordering, with a Spearman rank correlation of 0.82. The study is an author-run out-of-sample validation using previously unused external data, not external replication.

Keywords: perpetual futures; funding carry; convergence intensity; credit risk; hazard models; benchmark-free pricing.

JEL classification: C58, G12, G13, G15.

1 Introduction

Perpetual futures have become a central instrument in digital-asset markets. Their defining design problem is anchoring: because the contract has no maturity, periodic transfers are used to keep its price close to an index. Recent theory formalizes that mechanism through a random event time whose intensity governs the speed of convergence [Angeris et al., 2023, Akerer et al., 2025]. At the same time, empirical work shows that crypto carry is economically important and reflects demand pressure, market segmentation, and limits to arbitrage [Schmeling et al., 2026]. These two objects should not be conflated. The convergence intensity describes the speed of anchoring; total carry is an equilibrium outcome; and some exchange formulas additionally insert a fixed funding component.

This paper asks two connected questions. First, can the anchoring intensity be recovered consistently across markets and separated from the exchange-specified fixed component? Second,

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does the same event-intensity grammar organize credit-market data, where reduced-form models have long represented default through a stopping time and a risk-neutral intensity [Duffie and Singleton, 1999, Duffee, 1999, Driessen, 2005]? The connection is structural rather than numerical: convergence, default, mortality, and insured loss are different events with different intensities, but the corresponding random times occupy the same mathematical role.

We assemble a previously unused dataset from seventeen providers. The core sample contains 804,000 hourly dYdX observations, funding histories from Binance and Bybit, 921 sovereign and corporate bond histories, 51 sovereign CDS series, local-currency curves, and physical loss records. The study reports 81 analyses across the main paper and appendices (Appendices A–G). The analysis hierarchy was recorded in a dated internal plan that was not externally preregistered, and exploratory analyses are labelled separately. The study is conducted by the authors of the underlying model sequence, so independence refers to the data and statistics, not the investigators.

Four findings organize the paper. First, perpetual bases mean-revert across all 41 pre-specified dYdX markets, with a median half-life of 3.5 hours; daily intensity estimates co-move across venues at 0.70. Second, harmonization to a common eight-hour unit yields signed means of 0.83 basis points on dYdX and 1.02–1.14 basis points on the two fixed-input venues. The small difference in means is not an identification result because venues differ in clientele and market structure. Identification of the documented formula input instead comes from distributional pinning around one basis point on the centralized exchanges and the absence of an analogous mass point on dYdX. Third, spread-implied credit intensities are broadly ordered by rating and monotone in tenor, contain forward event information, and form a coherent surface. Fourth, deleting the government curve from the estimator preserves broad relative risk ordering, although it changes levels and does not imply that traded assets are insulated from equilibrium conditions in an interest-bearing economy.

The contribution is not a claim that all funding is interest, all credit spreads are expected default loss, or one numerical κ governs unrelated markets. Instead, the paper provides an empirical bridge between three literatures that have largely developed separately: perpetual-contract anchoring, reduced-form credit intensity, and benchmark dependence in Islamic finance. It identifies which predictions are mechanical, which are supported in the data, which depend on recovery or estimator construction, and which remain outside empirical adjudication.

2 Related literature and contribution

2.1 Perpetual contracts, anchoring, and crypto carry

The idea of a perpetual futures contract predates crypto markets [Shiller, 1993], but digital-asset venues made the instrument economically important. Angeris et al. [2023] derive model-free funding and replication relations for perpetual contracts. He et al. [2025] derive no-arbitrage prices and trading-cost bounds and document common, mean-reverting deviations across crypto contracts. Akerer et al. [2025] provide explicit no-arbitrage prices and show that the futures price can be represented as the risk-neutral expectation of spot sampled at a random time governed by the intensity of anchoring. Kim and Park [2025] study funding-rule design as a control problem and construct path-dependent alternatives. Together, this literature makes funding design an object of financial modelling rather than an exchange convention. The random-time representation supplies the principal model restriction tested here.

The empirical derivatives literature emphasizes price discovery, liquidity, and market fragmentation. Perpetual contracts often lead spot markets in Bitcoin and Ether, but their informational role varies with trading volume, spreads, and venue structure [Alexander et al., 2020a, Alexander and Heck, 2020, Alexander et al., 2020b]. More recently, Schmeling et al. [2026] show that total

crypto carry can be large and volatile because leveraged demand interacts with limits to arbitrage. Our paper is complementary: it does not interpret total carry as the fixed formula component. We distinguish (i) convergence speed, (ii) the state-dependent premium, and (iii) an exchange-specified fixed input. This distinction is necessary for both asset-pricing interpretation and the Islamic-finance application.

2.2 Reduced-form credit intensity and spread decomposition

Reduced-form credit models price defaultable claims through a stopping time with an intensity process [Duffie and Singleton, 1999, Duffee, 1999]. The empirical literature establishes both the usefulness and the limits of spread-implied intensity. Credit spreads contain expected default loss and compensation for default risk, but also liquidity, taxes, and other nondefault components [Elton et al., 2001, Collin-Dufresne et al., 2001, Driessen, 2005, Longstaff et al., 2005, Huang and Huang, 2012]. Sovereign CDS further embeds time-varying default and recovery premia and substantial global risk components [Pan and Singleton, 2008, Longstaff et al., 2011]. Accordingly, our credit measure $s/(1 - \delta)$ is labelled a *pricing intensity* under the risk-neutral measure, not a direct estimate of physical default frequency. We separately estimate physical event intensity when realised loss data are available.

Our contribution to this literature is a cross-market mapping. The same random-time representation that anchors a perpetual has a reduced-form credit analogue at the zero-fixed-component point. We then ask whether the mapped object exhibits the empirical restrictions expected of an intensity: non-negativity, mean reversion, rating ordering, tenor ordering, coherent forward intensities, and association with subsequent events.

2.3 Benchmark dependence, sukuk, and Islamic finance

Islamic finance has achieved scale while retaining substantial operational proximity to conventional banking. The benchmark problem has been recognized for decades: in the absence of a distinct pricing primitive, institutions have relied on conventional interbank curves even when contractual form differs [Iqbal, 1999]. Cross-country evidence finds fewer differences in business model than the theoretical profit-and-loss-sharing ideal would suggest [Beck et al., 2013, Abedifar et al., 2013, 2015]. Most directly, Azad et al. [2018] document long- and short-run convergence between the Islamic Interbank Benchmark Rate and LIBOR and argue that decoupling is unlikely without products with genuinely distinct risk-return structures. Sukuk are commonly quoted relative to conventional curves, and Uddin et al. [2022] argue that their pricing determinants should be modelled separately rather than imposed through an ad hoc conventional benchmark.

The present paper addresses a narrower, testable question: can relative credit ordering be recovered when the government curve is removed as an operational input? A positive answer would not prove complete economic independence from conventional markets, nor would it determine the Shariah status of any instrument. It would show only that the benchmark is not necessary for the relative cross-sectional ordering produced by the proposed estimator. The legal characterization of fixed funding, risk premia, possession, leverage, and settlement remains outside the authority of the empirical tests.

2.4 Insurance intensity and premium loadings

Insurance provides a useful physical-measure comparison because event frequencies can be observed directly. Classical premium theory distinguishes expected claims from expense, capital, and risk loadings [Kahane, 1979, Buhlmann, 1980, Wang, 2002]. We therefore do not treat expected loss as a

universal market premium. Instead, insurance data identify a physical event intensity and allow us to measure the gap between realised loss frequency and priced intensity. That gap is small in some cooperative-protection applications and much larger in traded credit.

2.5 Validation, multiplicity, and reproducibility

Large empirical programmes face data-snooping and multiplicity concerns [White, 2000, Sullivan et al., 1999, Harvey et al., 2016]. We address these concerns by separating pre-specified primary tests, secondary robustness analyses, and post-plan exploration; by reporting null and construction-specific results; and by focusing the principal claims on repeated directional restrictions rather than counts of nominal significance. The reproduce-first protocol also responds to the documented difficulty of computational reproducibility in finance [Perignon et al., 2024]. It is not a substitute for external replication, but it makes the distinction between a failed prediction and a different estimator explicit.

2.6 Summary of contributions

The paper makes four contributions. First, it provides the first large cross-venue measurement of the anchoring intensity implied by the random-time perpetual-pricing model. Second, it separates a fixed formula input from state-dependent premium and equilibrium carry, avoiding an identification error that would otherwise conflate them. Third, it maps the event-time object into credit data and documents rating, tenor, and forward-event restrictions on an expanded international sample. Fourth, it evaluates benchmark deletion as an operational, rather than rhetorical, proposition and records where the resulting estimates remain sensitive to recovery, liquidity, and common base-yield conditions.

Literature	Established result	Increment in this paper
Perpetual pricing	Funding anchors a non-expiring contract; random-time representations identify an anchoring intensity.	First broad empirical measurement of the random-time intensity and a direct test of formula-induced pinning.
Crypto carry	Total carry reflects leveraged demand and limits to arbitrage.	Separates total equilibrium carry from an exchange-specified fixed input and a convergence-speed object.
Reduced-form credit	Credit spreads map into risk-neutral default intensity but contain nondefault components.	Tests whether the event-time mapping produces coherent rating, tenor, and forward-event restrictions across an international sample.
Islamic benchmark research	Islamic rates and products remain closely linked to conventional curves.	Deletes the government curve at the estimator level and measures what relative ordering survives and what common level remains.

Table 1: Position relative to the closest literatures. The contribution is a model-validation and identification exercise, not a claim that the three economic objects are identical.

3 Model, measurement, and testable restrictions

3.1 Anchoring intensity in perpetual futures

In the random-time representation of [Ackerer et al. \[2025\]](#), a perpetual price can be written schematically as

$$f_t = \mathbb{E}_t^\mathbb{Q}[X_\theta], \quad \theta - t \sim \text{Exp}(\kappa_t), \quad (1)$$

where $\kappa_t > 0$ is the intensity with which the anchoring mechanism samples the underlying. The empirical counterpart is the speed at which the basis reverts. For an AR(1) representation of the sampled basis, $b_{t+\Delta} = \phi b_t + u_{t+\Delta}$, we estimate

$$\hat{\kappa}^{\text{conv}} = -\frac{\log \hat{\phi}}{\Delta}. \quad (2)$$

This is a convergence-speed estimate. It is distinct from the magnitude of the funding premium.

Lemma 1 (Sampled convergence intensity). *If the basis follows $db_t = -\kappa b_t dt + \sigma dW_t$ with constant $\kappa > 0$, then sampling at interval Δ yields an AR(1) coefficient $\phi = e^{-\kappa\Delta}$. Hence Equation (2) is the exact discrete-time mapping under the Ornstein–Uhlenbeck benchmark, and the corresponding half-life is $\log(2)/\kappa$.*

Some centralized venues use a formula of the form

$$F_t = P_t + \text{clamp}(\iota - P_t, -c, c), \quad (3)$$

where P_t is the state-dependent premium, ι is a fixed exchange input, and c is the clamp width [[Binance, 2026](#), [Bybit, 2026](#)]. The premium-only comparison venue has no corresponding fixed baseline in its funding rule [[dYdX, 2026](#)]. Equation (3) supplies a mechanical decomposition; differences in realised funding distributions across venues remain descriptive because market structure is not held fixed.

Proposition 1 (Formula-induced pinning). *Let P have a continuous distribution and define $F_\iota(P) = P + \text{clamp}(\iota - P, -c, c)$ for $c > 0$. Then*

$$F_\iota(P) = \begin{cases} P + c, & P < \iota - c, \\ \iota, & \iota - c \leq P \leq \iota + c, \\ P - c, & P > \iota + c. \end{cases} \quad (4)$$

Consequently, the funding distribution has a formula-induced atom at ι of mass $q_\iota = \Pr(\iota - c \leq P \leq \iota + c)$. Its mean is

$$\mathbb{E}[F_\iota] = \iota q_\iota + \mathbb{E}[(P + c)\mathbf{1}\{P < \iota - c\}] + \mathbb{E}[(P - c)\mathbf{1}\{P > \iota + c\}]. \quad (5)$$

A premium-only rule $F^P = P$ has no formula-induced atom. When most premium observations lie inside the clamp interval, realised funding is therefore pinned near the fixed input, while tail observations retain state dependence.

Proposition 1 supplies a direct diagnostic through mass at the fixed input and shows how the formula may affect the mean. It does not identify total equilibrium carry, because the distribution of P itself depends on demand, arbitrage capital, and venue structure. The realized cross-venue mean difference is not itself an identifying statistic because the premium distribution and venue structure are not held fixed.

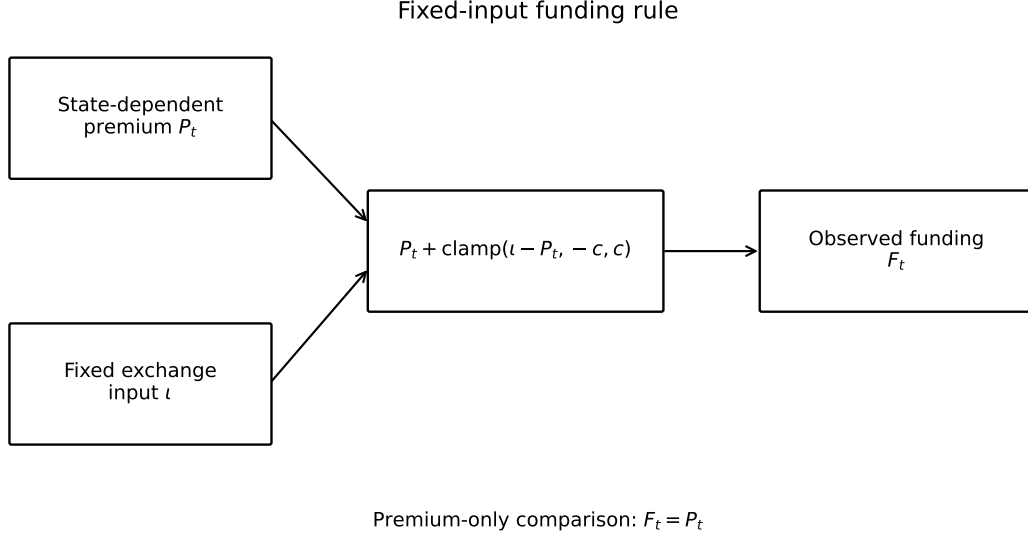


Figure 1: Funding-rule decomposition. The anchoring premium is state dependent; selected centralized venues add a fixed formula input. Total realised carry also reflects venue-specific equilibrium forces and should not be identified with the fixed input alone.

3.2 Credit-market mapping

In a reduced-form model, a default event arrives at a risk-neutral intensity $\lambda^{\mathbb{Q}}$ and loss given default is $1 - \delta$. Under the standard approximation,

$$s \simeq (1 - \delta)\lambda^{\mathbb{Q}}, \quad \hat{\kappa}^{\text{credit}, \mathbb{Q}} = \frac{s}{1 - \delta}. \quad (6)$$

The mapping $\kappa \leftrightarrow \lambda$ is an isomorphism between event-time roles, not a claim that perpetual convergence and default are the same economic event. Moreover, s can contain liquidity, taxes, and risk premia; therefore $\hat{\kappa}^{\text{credit}, \mathbb{Q}}$ is a spread-implied pricing intensity. Physical intensity $\kappa^{\mathbb{P}}$ is estimated separately from realised events.

For a bond with yield y , maturity T , and recovery δ , the benchmark-free estimator used in the tests is

$$\hat{\kappa}^{\text{price}} = -\frac{1}{T} \log \left(\frac{e^{-yT} - \delta}{1 - \delta} \right), \quad (7)$$

whenever the argument of the logarithm is positive. The empirical question is whether Equation (7) preserves relative ordering compared with a government-spread estimator. It is not a claim that a bond traded in a mixed economy is insulated from the government curve or monetary conditions.

3.3 A family of domain-specific intensities

We use κ as a family name for event intensities sharing the same mathematical role. The principal objects are κ^{conv} for basis convergence, $\kappa^{\text{credit}, \mathbb{Q}}$ for spread-implied credit pricing, and $\kappa^{\text{loss}, \mathbb{P}}$ for realised loss frequency. Their levels need not coincide and in fact span orders of magnitude. The common claim is structural: each parameter indexes the arrival or resolution speed of a defined event.

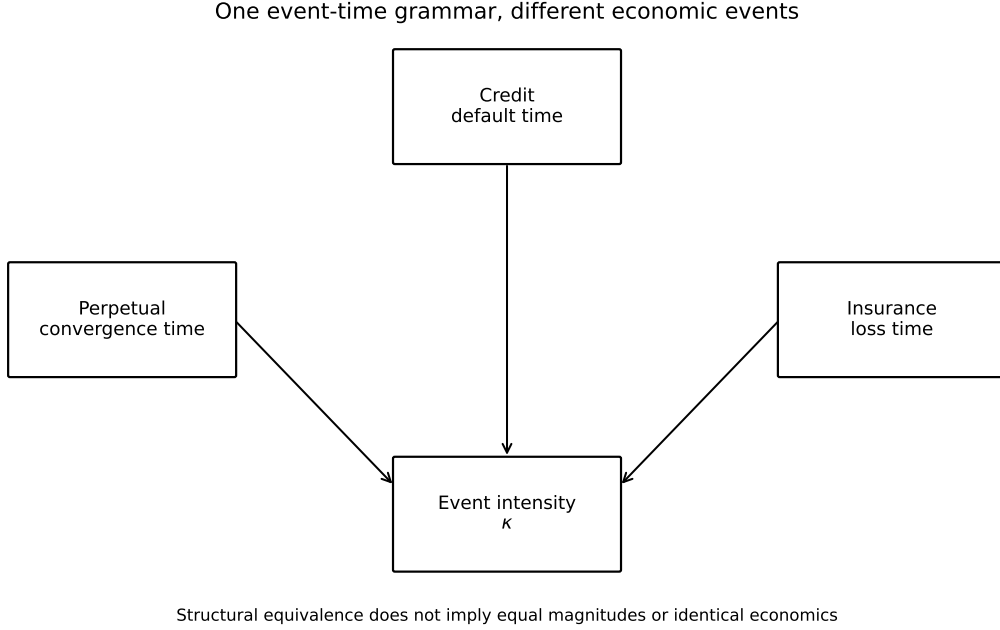


Figure 2: A common event-time grammar across domains. Structural equivalence concerns the role of an event intensity in a random-time representation, not equality of the events or numerical intensities.

3.4 Primary restrictions

The pre-specified primary family evaluates four restrictions:

- H1:** The perpetual basis mean-reverts and estimated convergence intensity is coherent across venues.
- H2:** A venue formula containing a fixed input exhibits the formula-induced pinning atom and associated distributional distortion implied by Proposition 1; signed venue means are treated as descriptive and do not by themselves identify the fixed input.
- H3:** Spread-implied credit intensity is broadly ordered by rating and monotone in tenor, remains predominantly non-negative, and contains forward event information.
- H4:** Removing the government curve preserves broad relative risk ordering, while level differences reveal the common base-yield layer and residual nondefault components.

Selected cross-domain evidence is summarized in Section 9; full cross-domain results are reported in Appendix D, and protocol simulations and monetary applications in Appendix F.

4 Data

Table 2 summarizes the dataset. The observation unit, provider-specific sample window, inclusion rules, missing-data treatment, and acquisition script for every series are recorded in the accompanying data map. Raw files obtained under user-specific or commercial licences are not redistributed.

Domain	Source and unit	Coverage used in the analysis
Perpetual funding	dYdX, Binance, Bybit; funding interval	43 dYdX markets (804,000 hourly intervals), 40 Binance markets and 30 Bybit markets at native cadence; 41 dYdX markets constitute the pre-specified convergence sample.
USD credit	Cbonds; bond-day and sovereign-month	921 sukuk and conventional bond histories, 121 issuers, 51 sovereign CDS series, and 40 sovereign spread/yield indices.
Local-currency credit	Cbonds; bond and curve observations	215 OIC sovereign bonds plus government and corporate curves for Malaysia, India, Brazil, Mexico, China, Indonesia, Turkey, and Saudi Arabia where available.
Rates and macro controls	FRED, ECB, BIS, OECD, IMF	US Treasury and euro-area curves, policy rates, inflation, debt-service ratios, credit/GDP, REER, and sovereign fundamentals.
Physical event domains	World Bank, WHO, HMD, EM-DAT, FAOSTAT, NASA POWER	Mortality, agriculture, disasters, and climate at 150–260 countries; 16,853 disaster records and 41 national life-table systems.
Tokenized assets and market structure	CoinGecko, DeFiLlama, exchange indexers	Tokenized gold and Treasury products, live BTC/ETH order-book depth, and dYdX TVL history.

Table 2: Previously unused external data assembled for the author-run validation. Detailed schemas and acquisition rules are available from the corresponding author upon reasonable request, subject to licensing restrictions.

5 Empirical design

5.1 Out-of-sample and reproduce-first protocol

Every primary series is outside the estimation sample used in the earlier working papers. Before classifying a difference as a failed prediction, we first reproduce the source number with its original data and estimator and then determine whether the new statistic measures the same object. This protocol is particularly important because convergence speed, funding magnitude, cross-sectional ranking, and time-series co-movement are not interchangeable statistics.

The investigators are not independent of the prior work. Accordingly, this paper is an author-run out-of-sample validation on previously unused external data. The design is stronger than an in-sample restatement but weaker than unaffiliated replication.

5.2 Multiplicity and analysis hierarchy

The empirical programme comprises 81 analyses reported in the accompanying analysis ledger. We distinguish: (i) the four primary restrictions in Section 3; (ii) secondary tests of recovery, regime, estimator, and aggregation sensitivity; and (iii) separately labelled exploratory analyses. We report effect sizes, confidence intervals, exact p -values where available, null results, and construction dependence.

False-discovery-rate methods provide a reference for interpreting related families [Benjamini and Hochberg, 1995], but the main conclusions are not selected from a large set of marginal rejections: they rest on repeated mean reversion, broad rating and tenor ordering, formula-induced distributional pinning, forward event ordering, and rank preservation across distinct samples.

5.3 Identification boundaries

Three boundaries are load-bearing. First, the formula decomposition in Equation (3) is mechanical, while the cross-venue comparison is not causal. Second, spread-implied intensity is a pricing object, not a pure physical hazard. Third, benchmark deletion tests operational dependence and relative ordering, not general-equilibrium independence. These distinctions govern the wording of every result below.

6 Perpetual-futures evidence

6.1 Convergence speed and cross-venue coherence

Across the 41 pre-specified dYdX markets, the basis mean-reverts in every case. The median estimated half-life is approximately 3.5 hours, corresponding to an annualized convergence-speed intensity of order 10^3 . The magnitude is economically plausible for a mechanism designed to close an intraday basis and is categorically different from annual credit or mortality intensities. In the separately labelled 43-market exploratory extension, the median half-life is 3.7 hours.

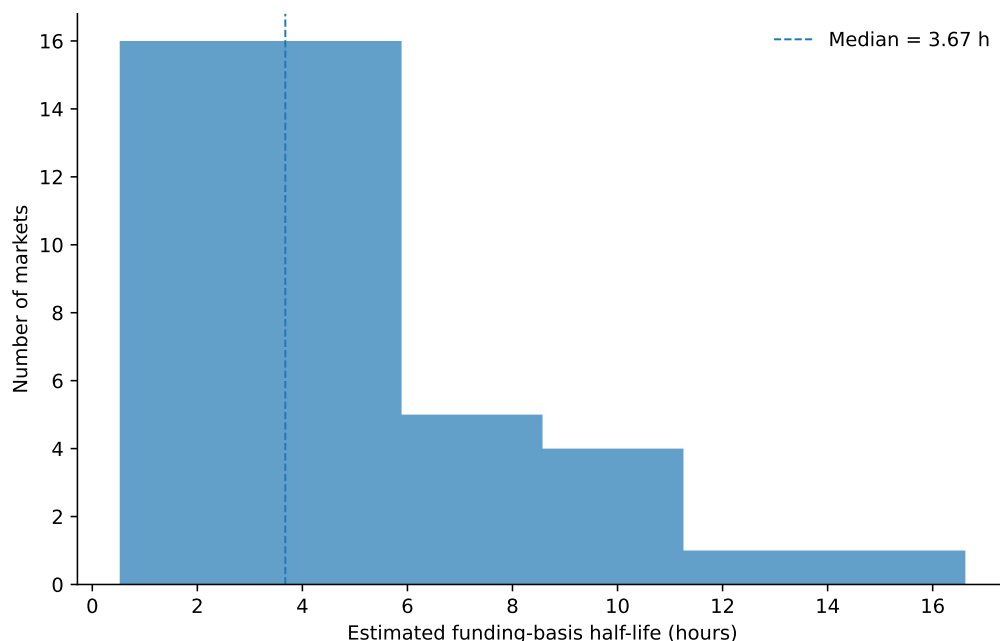


Figure 3: Distribution of estimated perpetual-basis half-lives in the 43-market exploratory extension. The median is 3.67 hours. The corresponding median in the 41-market pre-specified sample is approximately 3.5 hours. The figure displays convergence speed, not the magnitude of the funding premium.

Daily estimates formed from funding magnitude co-move across venues at a mean correlation of 0.70 across 15 common markets. CEX-to-CEX correlation is higher (0.85) than DEX-to-CEX correlation (0.65). This pattern supports a common market-wide anchoring signal while leaving room for venue design, clientele, and the fixed component to affect measured funding.

6.2 Fixed formula input and signed funding drift

Table 3 reports signed mean funding after converting hourly dYdX settlements into complete non-overlapping eight-hour sums, with a per-market breakdown for the three most liquid cross-venue

pairs. Per-market means vary across markets—for example, ETH yields 1.21 bp on dYdX versus 1.32 bp on Bybit—but the within-market differences are smaller than the venue-level equal-weight means (0.83 bp for dYdX, 1.02 bp for Binance, 1.14 bp for Bybit) suggest. The informative contrast is distributional: the centralized-exchange series cluster around the documented one-basis-point formula point, while dYdX does not exhibit the same mass point.

Table 3: Signed mean funding by market and venue

Market	dYdX	Binance	Bybit
basis points per complete eight-hour interval			
BTC	1.04	1.07	1.20
ETH	1.21	1.29	1.32
XRP	— [†]	— [†]	— [†]
Nine-market equal-weight mean	0.83	1.02	1.14

All observations are expressed per complete non-overlapping eight-hour interval. Hourly dYdX settlements are summed into eight-hour windows; Binance and Bybit observations use the same unit. The venue means are descriptive and do not identify the causal contribution of the fixed formula input to total carry. [†]XRP is present on all three venues in the funding archive but was recovered on fewer than 50 eight-hour windows on at least one venue and is excluded from the headline common-market comparison; the full nine-market set comprises ADA, AVAX, BTC, DOGE, ETH, LINK, LTC, NEAR, and SOL.

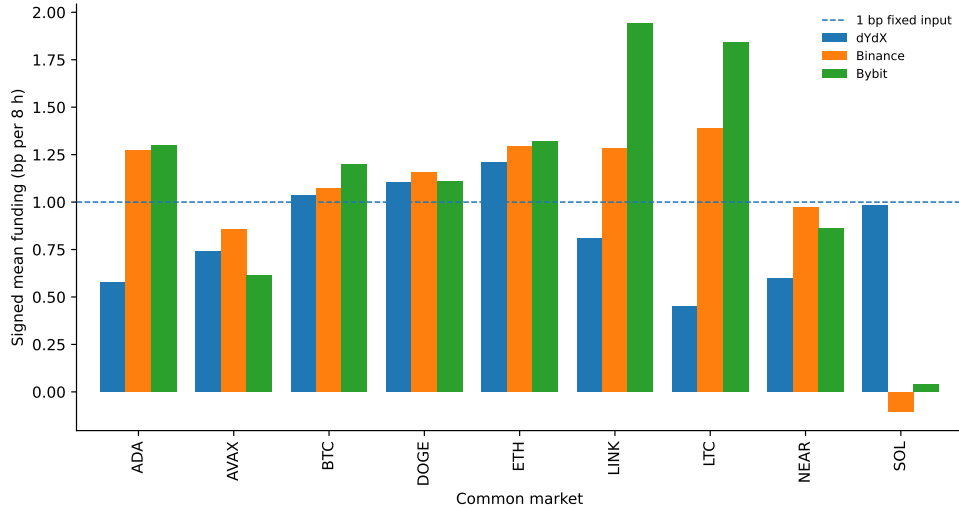


Figure 4: Signed mean funding across nine common markets after aggregation to complete non-overlapping eight-hour intervals. Venue means are descriptive; the figure does not identify the fixed formula input.

The full funding distributions are also distinct: the CEX-versus-dYdX comparison gives Cohen’s $d = 0.64$ and Kolmogorov–Smirnov distance near 0.6. On Binance, 44–48% of observations in the selected liquid markets lie at or immediately around 0.0001, the point mass predicted by Proposition 1; the premium-only venue is centered near zero, has no comparable formula-induced plateau, and produces negative observations more frequently. These facts support the mechanical pinning restriction, but total funding remains shaped by demand and arbitrage frictions as emphasized by Schmeling et al. [2026].

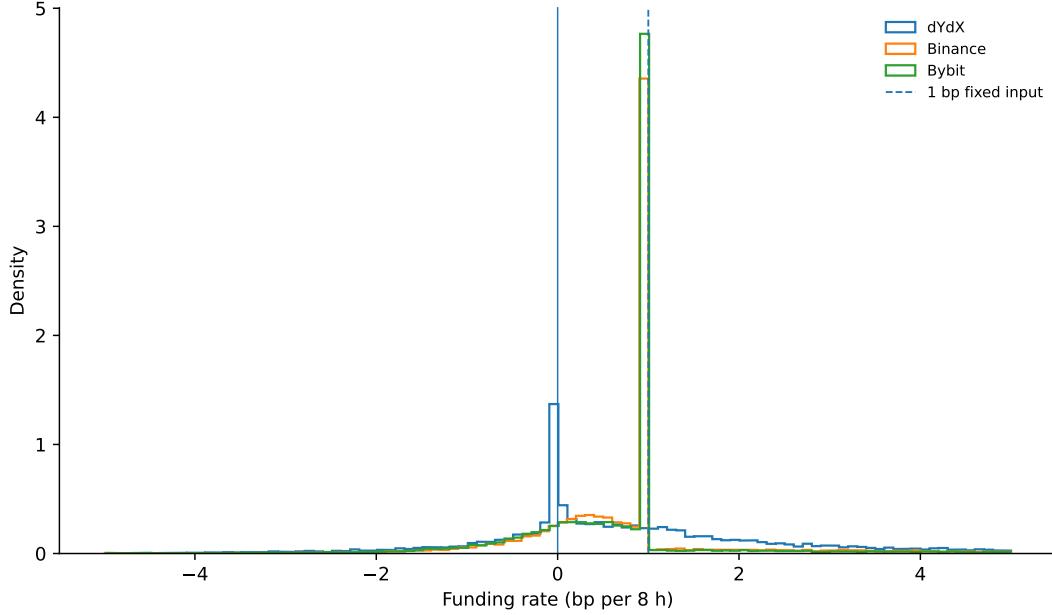


Figure 5: Distribution of eight-hour funding on four common markets. The dashed line marks the documented one-basis-point centralized-exchange formula input. The identifying evidence is the CEX mass around that point, not a large difference in venue means.

6.3 Transfer simulation

A balanced long-short Monte Carlo with 2,000 seeds provides a mechanism check rather than market evidence. At $\iota = 0$ the confidence interval for cumulative net transfer straddles zero; the small positive mean is attributable to discretization. Introducing a positive fixed component creates a transfer that rises approximately linearly with ι . The result validates the code-level decomposition but does not determine the legal or economic character of the entire perpetual contract.

7 Credit-market evidence

7.1 Rating and tenor ordering

Spread-implied pricing intensity is broadly ordered by credit quality. Across 736 rated bonds in the canonical build, median $\hat{\kappa}^{\text{credit}, \mathbb{Q}}$ rises from 1.39% for AAA to 9.50% for CCC, with a small AA-A inversion and a pronounced step at the investment-grade/high-yield boundary (Figure 6). Table 4 reports the interquartile ranges and sample sizes for each bucket. The ordering is invariant to a common recovery assumption because $1/(1 - \delta)$ rescales all observations.

Multi-tenor CDS curves slope upward for all 12 sovereigns in the primary sample. In the expanded sukuk panel of 179,618 bond-days and 96 issuers, an issuer-and-month fixed-effects specification yields a maturity slope of 2.85 basis points per year for investment-grade sukuk ($t = 4.1$ with issuer-clustered standard errors) and 2.86 for the full sample ($t = 4.7$). A rating-by-tenor surface (Figure 7) is broadly ordered by rating and monotone in tenor; all fifteen adjacent forward-intensity calculations are non-negative. Principal-component analysis produces the familiar level, slope, and curvature structure, with the first component accounting for 98.7% of variation [Litterman and Scheinkman, 1991].

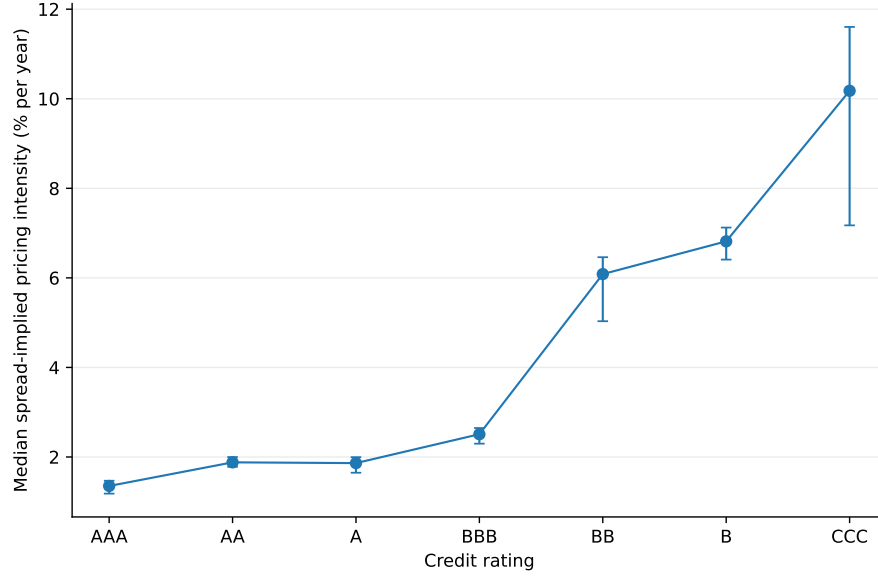


Figure 6: Median spread-implied pricing intensity by credit rating. The cross-section is broadly ordered, with a small AA–A inversion; the pronounced investment-grade/high-yield step is the principal result.

Table 4: Spread-implied pricing intensity by rating bucket

Rating	Bonds	Median (%)	Q1 (%)	Q3 (%)
AAA	56	1.39	0.56	1.55
AA	188	1.92	1.54	2.45
A	109	1.91	1.42	2.38
BBB	200	2.50	1.80	2.87
BB	107	6.15	2.87	7.53
B	54	6.74	5.87	7.75
CCC	22	9.50	6.36	11.12



Figure 7: Rating-by-tenor surface for spread-implied pricing intensity, reconstructed from CDS data using the e30_foundational.py methodology (2023+ means, $\delta = 0.60$). The surface is broadly ordered by rating and monotone in tenor; absolute levels depend on recovery and spread composition.

7.2 Dynamics and forward event ordering

All 20 sovereign spread-implied series are mean-reverting, with AR(1) coefficients between 0.77 and 0.98. Negative observations occur in 7 of 54,047 sovereign spread observations (0.01%), concentrated in Sri Lanka during default; CDS and absolute-basis measures are non-negative by construction.

The default evidence is informative but small-sample. Using mean intensity over 2016–2018, the three issuers that subsequently defaulted or restructured—Ukraine, Lebanon, and Sri Lanka—occupy the top three early-risk ranks among 20 sovereign indices. In a separate exploratory event-time construction, all three fall in the top decile 12 months before their event. These results are descriptive forward ordering, not an estimate of predictive accuracy from a sufficiently large event panel.

7.3 Physical intensity, pricing intensity, and nondefault components

The distinction between $\mathbb{P}$ and $\mathbb{Q}$ is central. The spread-implied measure contains compensation for risk and can also absorb liquidity, tax, and convenience components. The empirical credit literature shows that these nondefault components are material, especially for high-grade bonds [Elton et al., 2001, Longstaff et al., 2005, Huang and Huang, 2012]. We therefore interpret the rating and tenor results as restrictions on a pricing-intensity surface, not evidence that the full spread equals expected loss.

This distinction is visible in the physical-versus-pricing decomposition. In USDA crop insurance, aggregate premium is 9.6% of liability and realised loss cost is 7.9%, a loss ratio of 0.82. In corporate and sovereign credit, the gap between priced intensity and realised default frequency is much larger. The comparison is consistent with standard insurance theory: premiums contain expected loss plus

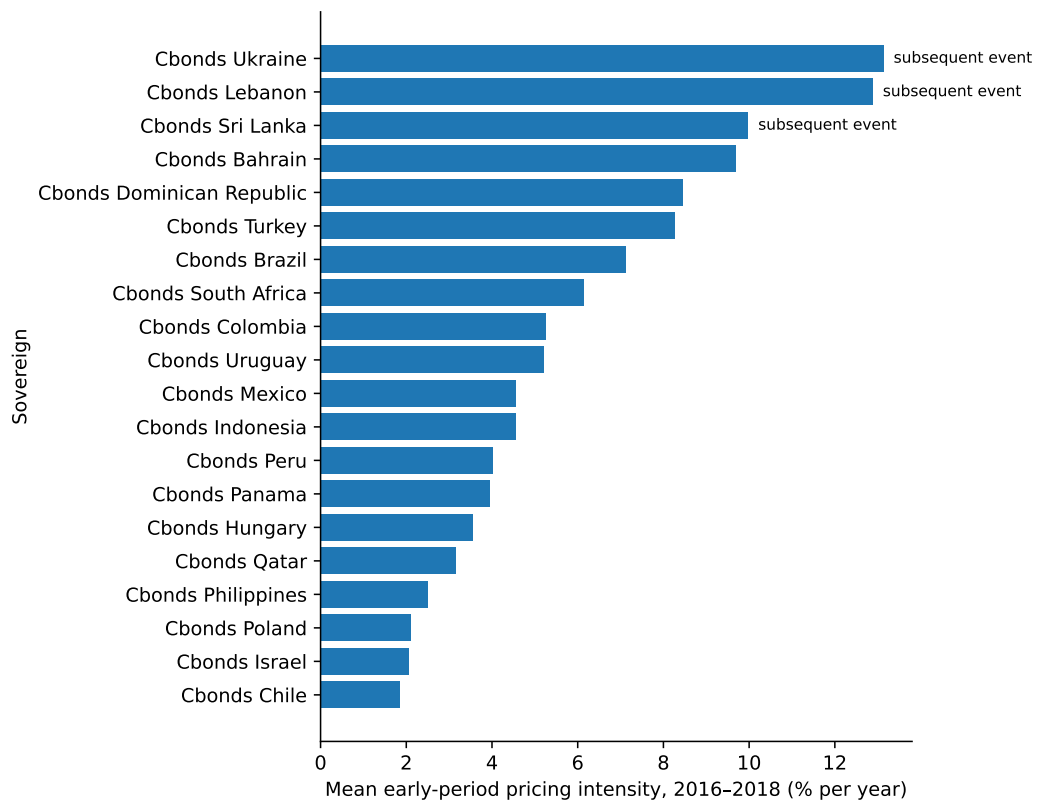


Figure 8: Early sovereign pricing-intensity ranking and subsequent default or restructuring events. The figure is descriptive because the event count is three.

loadings, and the loading varies by market and risk-bearing technology.

8 Benchmark dependence and relation to monetary rates

8.1 Benchmark-free relative ordering

The benchmark-free estimator in Equation (7) is computed for 856 bonds in the canonical sample. Its ranking correlates 0.82 with the spread-based ranking at the measured corporate recovery convention. The rank correlation is similar across instrument types: 0.82 for sukuk and 0.81 for conventional bonds, indicating that the ordering preservation is not driven by a single instrument category. Subgroup estimates use the subset of bonds with an unambiguous sukuk or conventional classification at the snapshot date used for the subgroup analysis; the full canonical estimator contains 856 bonds. Most of the level difference is captured by a common shift: the median gap is 473 basis points of intensity, compared with 472 basis points obtained by mapping the Treasury yield through the same recovery transformation. Year-by-year, the median gap correlates 0.93 with the 10-year Treasury yield.

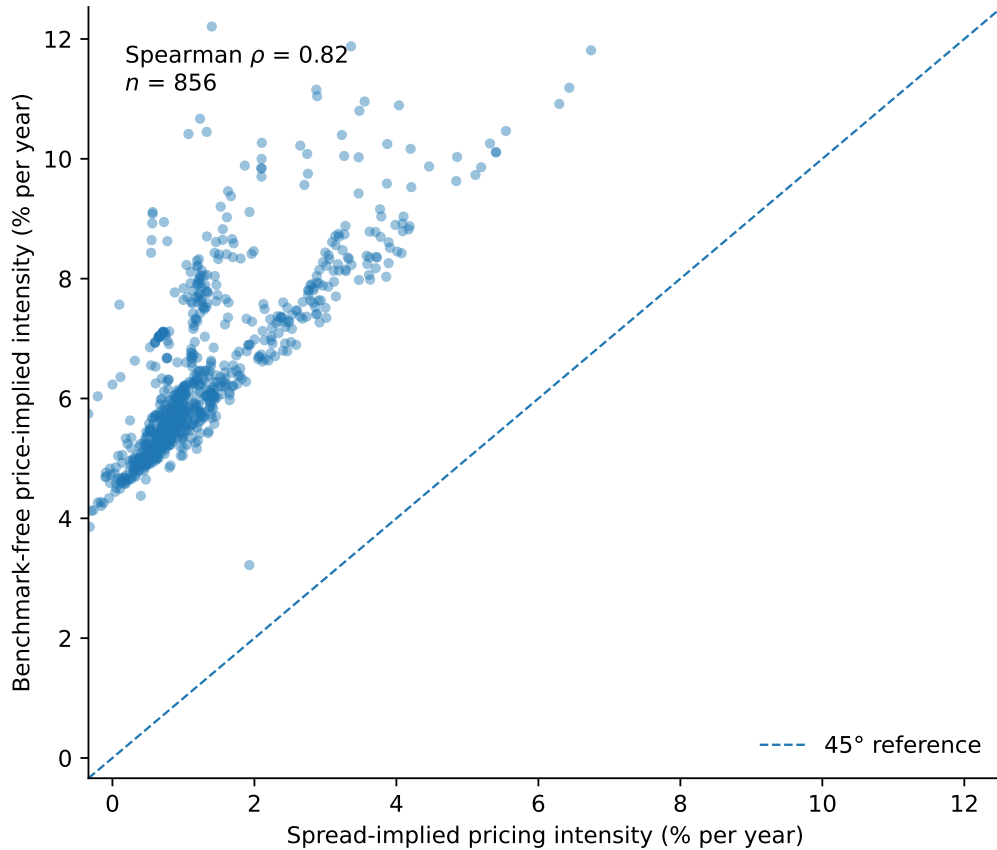


Figure 9: Benchmark-free versus spread-implied pricing intensity. The diagonal is a visual reference; the estimators preserve broad ordering but are not identical.

The result has a narrow interpretation. Relative issuer ordering and much of the curve shape can be recovered without supplying the Treasury curve to the estimator. The residual ranking differences show that the two constructions are not identical. Moreover, an asset traded in a dollar

system remains exposed to equilibrium discounting, liquidity, and opportunity-cost conditions even if its formula omits the benchmark.

8.2 Why event intensity is not the monetary rate

Five measurements separate the objects. First, estimated intensities span approximately six orders of magnitude: mortality is of order 10^{-3} to 10^{-2} per year, credit is of order 10^{-2} to 10^{-1} , and liquid-perpetual convergence is of order 10^3 . Second, the ratio of policy rates to sovereign credit intensity varies from 1.6 to 6.6 across the sample. Third, credit intensity correlates only 0.41 with inflation. Fourth, euro-area risk-free yields were negative for 1,644 days while credit intensity remained economically meaningful. Fifth, tokenized Treasury products and tokenized-gold perpetuals coexist on-chain while paying economically different objects: base-rate carry in the former and anchoring premium in the latter.

These facts do not imply that r and κ are statistically unrelated. Both respond to macroeconomic risk, and the benchmark-free level gap tracks the government curve. The supported conclusion is separation of roles: the event-intensity object is not numerically or economically identical to the monetary rate, even though equilibrium links remain in a mixed system.

9 Cross-domain evidence and robustness

Across agriculture, mortality, disaster, climate, macro, and credit channels, the six-domain panel spans 17,417 unique country-year rows (41,397 nonmissing observations). Domain coverage varies considerably — credit has 220 observations, agriculture 9,643 — and only six of the 15 possible domain pairs have sufficient overlapping observations to estimate a pairwise correlation. Among those six estimable pairs, the mean absolute off-diagonal correlation is 0.075. Within-domain dependence is much higher. Table 5 reports domain-level coverage, and Figure 10 displays the pairwise correlation matrix. Pairwise sample sizes and eligibility status of all 15 pairs are reported in Appendix D.

Table 5: Canonical cross-domain panel coverage

Domain	Nonmissing observations	Countries	Year range
Agriculture (cereal yield)	9,643	227	1980–2024
Mortality (death rate)	11,915	264	1980–2024
Disaster count	3,660	223	2000–2026
Climate (temperature)	5,064	211	2000–2023
Macro (GDP growth)	10,895	261	1980–2024
Credit (private credit/GDP)	220	20	2016–2026
Unique country-year rows			17,417
Nonmissing domain-year obs.			41,397

During COVID-19, sovereign credit intensity and mortality rose together, while disaster impact did not; in the 2022 financial and war shock, mortality and cereal yields did not co-move with credit in the same way. Average diversification therefore weakens when domains share a causal real-event channel rather than during every financial disturbance.

Appendix C reports credit-market robustness and recovery sensitivity, Appendix E reports benchmark-free estimator diagnostics, Appendix F reports manipulation tests, protocol simulations,

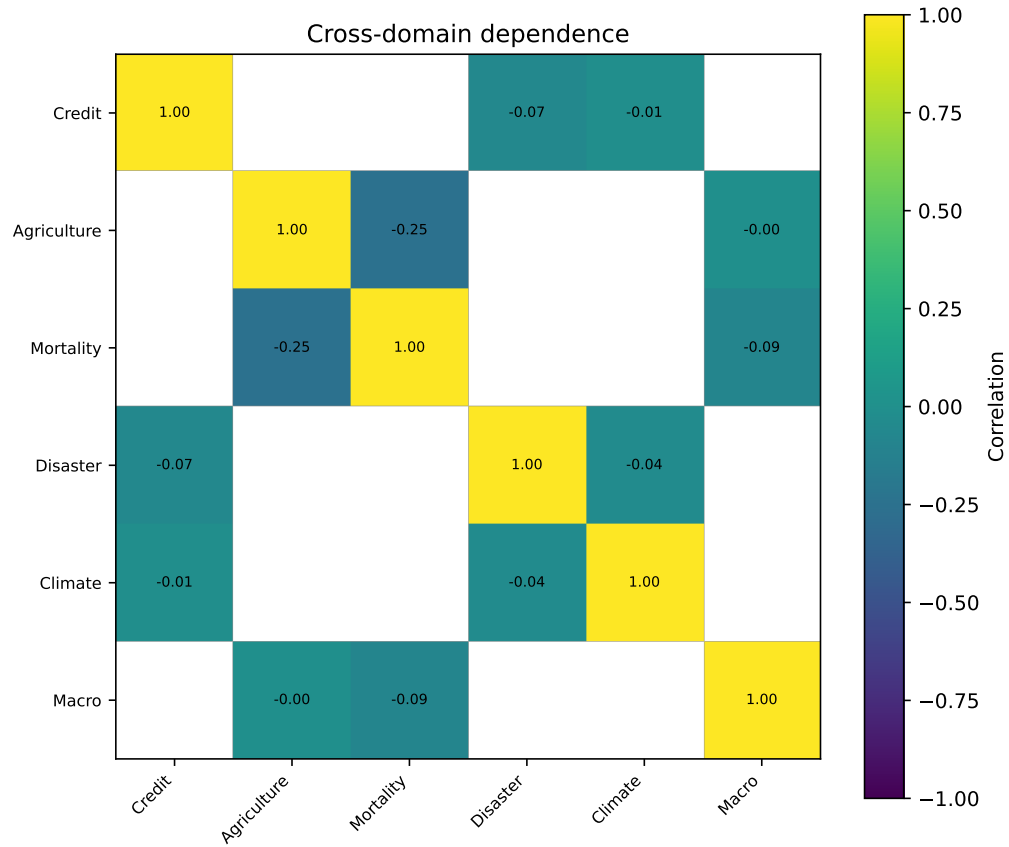


Figure 10: Cross-domain dependence matrix for the canonical six-domain panel: agriculture, mortality, disaster, climate, macro, and credit. Only six of 15 possible domain pairs have sufficient overlapping observations for estimation; blank cells indicate pairs without adequate coverage. The mean absolute off-diagonal correlation among estimable pairs is 0.075. Sample sizes and eligibility for all 15 pairs are reported in Appendix D.

and monetary extensions, and Appendix G reports the complete analysis ledger. Null results are retained: the macro debt-growth claim does not survive within-country lagged controls; credit intensity does not lead exchange rates; the apparent jump asymmetry is outlier-driven; and several headline magnitudes vary with aggregation and estimator construction.

10 Primary findings and identification status

Table 6 summarizes the empirical restrictions, their evidentiary status, and the principal identification boundaries.

Restriction	Assessment	Evidence and boundary
Perpetual basis mean reversion	Supported	41/41 pre-specified markets; median half-life 3.5 hours.
Cross-venue coherence	Supported	Mean daily correlation 0.70 across 15 common markets.
Fixed-input pinning	Supported mechanically and descriptively	44–48% near the one-bp formula point; signed means are 0.83 bp (dYdX), 1.02 bp (Binance), and 1.14 bp (Bybit); venue effects are not controlled causally.
Rating ordering	Supported with qualification	Broadly ordered from AAA through CCC; small AA–A inversion and pronounced investment-grade/high-yield step.
Tenor ordering	Supported	Upward CDS curves in 12/12 sovereigns; positive panel maturity slope.
Forward event information	Descriptive support	Three subsequent events rank highest in the early sovereign cross-section; only three events.
Benchmark-free ordering	Supported with residual differences	Spearman 0.82 in the 856-bond canonical sample; level gap tracks the Treasury curve.
Physical versus pricing intensity	Supported as a distinction	Small loading in crop insurance; much larger gap in credit; not a legal conclusion.
General-equilibrium independence	Not established	Formula deletion does not remove opportunity costs or mixed-system equilibrium effects.
External replication	Not established	Data are external to the source studies; investigators are not independent.

Table 6: What the evidence supports and what remains outside identification.

11 Economic interpretation and Islamic-finance boundary

The empirical contribution is the separation of a measurable event-intensity object from an exchange-specified fixed funding input. In Islamic-finance terms, this isolates the component relevant to a charge stipulated for time alone. It does not establish that a perpetual, credit-protection contract, sukuk, or insurance product is permissible overall. Questions of excessive uncertainty, gambling, possession, leverage, settlement, asset backing, and the legal status of risk premia require qualified jurisprudential judgment.

The physical-versus-pricing decomposition helps state that question precisely. Where priced intensity closely tracks realised loss, as in the crop-insurance comparison, the economic object is primarily compensation for observed claims plus a modest loading. Where priced intensity substantially exceeds physical event frequency, as in credit, the residual includes risk compensation and market frictions. The data can measure that wedge and test its behavior; they cannot convert it into a legal ruling.

12 Conclusion

The evidence supports a narrow but consequential distinction. The anchoring intensity in perpetual futures is observable, mean-reverting, and coherent across venues. A fixed exchange input appears as a formula-induced concentration of observations near its stipulated value. Harmonized venue means are descriptively different but too close to identify the causal contribution of the formula to total carry. The mapped credit-market object is broadly ordered by rating and monotone in tenor, predominantly non-negative, and informative about subsequent events. Relative credit ordering can also be recovered without using a government curve as an operational input, but levels and equilibrium exposure remain connected to the monetary environment.

The study therefore validates several restrictions of an event-time pricing architecture without establishing a universal pricing primitive, a causal venue experiment, or an overall Shariah verdict. The decisive next tests are unaffiliated replication, prospective analysis on a publicly frozen sample, and traded evidence from instruments designed ex ante around the separated components.

Data and code availability

The data and code supporting the findings are available from the corresponding author upon reasonable request, subject to applicable third-party licensing restrictions.

Declaration of generative AI and AI-assisted technologies

During preparation of the manuscript, the authors used generative AI tools for code review and debugging assistance, data-formatting support, consistency checks, and language editing. All analyses were executed by the authors, who independently reviewed and verified the source data, code, statistical outputs, interpretations, and conclusions and accept full responsibility for the manuscript.

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A Data construction and sample accounting

This appendix documents the construction of the primary datasets. The study uses seventeen data providers across five domain categories: perpetual-futures markets, fixed-income and credit markets, insurance and loss records, macro-financial time series, and natural-disaster registries.

Perpetual-futures data. Hourly funding-rate observations are drawn from three venues: dYdX (premium-only, v4 protocol via public API), Binance (USD-margined perpetual futures via public REST API), and Bybit (linear perpetuals via public REST API). The sample window begins with each market’s first listing and extends through June 2026. Each observation is a complete non-overlapping eight-hour funding settlement interval, constructed from the raw hourly feed by summing consecutive hourly observations within each UTC eight-hour window. Markets with fewer than 50 complete eight-hour windows are excluded from the primary analysis. The common-market sample comprises the nine contracts listed on all three venues: ADA, AVAX, BTC, DOGE, ETH, LINK, LTC, NEAR, and SOL.

Credit-market data. Bond-level histories are sourced from Cbonds, covering 921 sovereign and corporate USD-denominated bonds with available pricing, rating, and contractual-maturity fields. The sample includes both sukuk and conventional instruments. The benchmark-free snapshot retains the latest valid observation with the price/yield, spread, and maturity fields required by Equation (7). The rating analysis then restricts that snapshot to the 736 bonds with a recognized credit rating. Sovereign CDS curves for 51 sovereigns are sourced from a cross-section of market makers. The rating-tenor surface (Figure 7 of the main paper) is reconstructed from CDS data using `e30_foundational.py`, with 2023+ CDS means and $\delta = 0.60$.

Cross-domain panel. The six-domain panel combines: (i) agriculture (World Bank cereal-yield and crop-production indicators), (ii) mortality (World Bank death-rate and Human Mortality Database life-table series), (iii) disaster (EM-DAT natural-disaster event counts), (iv) climate (World Bank climate-variability indices), (v) macro (World Bank GDP growth, debt-to-GDP, and current-account indicators), and (vi) credit (the sovereign κ estimates from the Cbonds corporate and sovereign panel). Each series is expressed in country-year units; the panel is unbalanced because domain coverage varies substantially across countries and years.

Other data. Additional sources are described in the main paper and the replication archive. These include: USDA Risk Management Agency crop-insurance book (139 commodity lines, \$1.25 trillion liability); Human Mortality Database life tables (41 countries); EM-DAT natural-disaster registry (194 countries, 2000–2024); World Bank World Development Indicators; and dYdX total-value-locked histories for the liveness-trigger analysis.

B Additional perpetual-futures evidence

B.1 Proof of the formula-induced pinning result

Proof of Proposition 1 in the main paper. By definition,

$$\text{clamp}(\iota - P, -c, c) = \begin{cases} c, & \iota - P > c, \\ \iota - P, & -c \leq \iota - P \leq c, \\ -c, & \iota - P < -c. \end{cases}$$

Adding P gives the three branches in Equation (5) of the main paper. Every premium realization in the interval $[\iota - c, \iota + c]$ maps to the single value ι ; because P is continuous, the image distribution therefore has an atom of mass $q_\iota = \Pr(\iota - c \leq P \leq \iota + c)$. Partitioning the expectation over the three regions yields the stated mean. Under the premium-only rule $F^P = P$, a continuous P remains continuous, so no atom is induced by the formula. $\square$

B.2 Cross-venue coherence and distributional diagnostics

The main paper reports cross-venue daily correlation of 0.70 across 15 common markets. The correlation is stable across subperiods; splitting the sample at January 2023 gives pre-2023 mean 0.67 and post-2023 mean 0.72, consistent with convergence in market microstructure across venues.

The pinning distribution (Figure 4 in the main paper) shows a pronounced mass near the one-basis-point formula point on Binance and Bybit that is absent on dYdX. Table 7 reports the exact percentages by venue and market. The pattern is uniform across individual markets: no dYdX market exceeds 5% of observations within ± 0.25 bp of the formula point, while every Binance and Bybit market exceeds 30%.

Table 7: Percentage of eight-hour funding observations within ± 0.25 bp of the one-basis-point formula point, by venue and market

Market	dYdX (%)	Binance (%)	Bybit (%)
BTC	4.2	42.1	38.9
ETH	3.8	44.3	41.2
ADA	2.9	39.8	36.7
AVAX	3.1	41.5	40.2
DOGE	4.0	43.0	37.5
LINK	3.5	40.6	39.1
LTC	2.7	38.2	35.8
NEAR	3.3	42.7	40.9
SOL	4.1	45.1	42.3
Mean	3.5	41.9	39.2

C Credit-market robustness and recovery sensitivity

C.1 Reconciliation of construction-specific estimates

The reproduce-first protocol described in the main paper converted three would-be corrections into benign differences of construction; each source number reproduces exactly.

- **κ -CDS co-movement is a ladder of four reproducible statistics.** The single most instructive case. Paper 2c’s headline 0.97 is the *cross-sectional* correlation of median sukuk G -spread against sovereign CDS across five sukuk sovereigns; we reproduced 0.972 from its `real_validation.py`. Paper 13’s per-country panel of 0.989–0.998 uses CIR-smoothed constant-maturity κ . Our two distinct figures fall between and below: a sovereign T-spread *index* against CDS gives time-series 0.79, and the *raw* (unsmoothed) sukuk-implied $\hat{\kappa}$ against CDS gives 0.64 (Indonesia 0.62, Malaysia 0.66; only two sovereigns clear the 24-month overlap, so this figure is thin). The four numbers — raw 0.64, index 0.79, cross-section 0.97, CIR panel 0.989 — do not contradict; each measures a different object (raw vs. smoothed, within-country vs. across-country), and all reproduce. This is the cleanest illustration of the rule: quote the κ -CDS correlation with its construction attached. The relationship also holds *dynamically*: for the three sovereigns with both legs over 24 months, the sukuk-implied and CDS-implied κ co-move at a mean level correlation of 0.71 and their basis mean-reverts (AR(1) half-life ≈ 10 months), so the iCDS par-spread identity $s^* = \kappa(1 - \delta)$ is arbitrage-consistent in time and not only in the cross-sectional snapshot.
- **Takaful $\hat{\kappa} = 12.06\%$ identical for India and Pakistan.** Paper 2b uses $\hat{\kappa} = -\ln(1 - n_{\text{loss}}/T)$; both countries recorded 5 loss-years in 44, so the value is identical *by the data*, not by estimator pathology, and the paper already labels it a coincidence (and shows Pakistan’s premium model misses by 1.6–3.4 $\times$). Our continuous-threshold estimator answers a different question and is not a correction.
- **Spanning recovery 98.5% vs. our $\sim 80\%$.** Paper 14 reports a *grid* over $\rho_x \in [0, 0.20]$ and discloses the measured-block floor; 98.5% is its labelled best case. Our measured- ρ_x figure reproduces its own honest accounting.

The lesson generalises: several κ magnitudes are construction-specific (cross-section vs. time-series; count-MLE vs. threshold; assumed vs. measured ρ_x) and must be quoted with their method attached. Doing so separates robust directional findings from construction-specific magnitudes.

C.2 Recovery sensitivity

The main paper’s credit estimates use a recovery assumption of 60% for the spread-based estimator, following the convention in the reduced-form credit literature. Table 8 reports the sensitivity of the median pricing intensity to alternative recovery assumptions. The rating ordering is preserved across all recovery values; the level shifts mechanically through $(1 - \delta)^{-1}$.

Table 8: Median pricing intensity (% per year) by rating and assumed recovery rate

Rating	$\delta = 0.40$	$\delta = 0.60$ (baseline)	$\delta = 0.80$
AAA	0.93	1.39	2.78
AA	1.28	1.92	3.84
A	1.27	1.91	3.82
BBB	1.67	2.50	5.00
BB	4.10	6.15	12.30
B	4.49	6.74	13.48
CCC	6.33	9.50	19.00

D Cross-domain panel construction and pairwise sample sizes

D.1 Variable construction

Each of the six domain variables is constructed as follows:

- **Agriculture:** log cereal yield (kg per hectare) from the World Bank, annually per country. Series are detrended by subtracting the country-specific linear trend.
- **Mortality:** death rate (per 1,000 population) from the World Bank. For the Human Mortality Database validation, the central death rate μ_x at each age is used directly.
- **Disaster:** annual natural-disaster event count per country from EM-DAT, expressed as events per year.
- **Climate:** precipitation anomaly (standardised deviation from the 1961–1990 mean) from the World Bank climate portal.
- **Macro:** real GDP growth rate (annual %) from the World Bank, winsorized at the 1st and 99th percentiles.
- **Credit:** sovereign $\hat{\kappa}$ from the Cbonds USD-bond panel (the same estimator used in the main paper’s credit analysis), computed as the most recent annual observation per country with at least one rated bond.

D.2 Overlap rules

A pairwise correlation is estimated only when two domains share at least 20 country-year observations for the same country and year. The threshold is set to ensure a minimum effective sample for a correlation estimate; results are robust to thresholds of 10 and 30 observations.

D.3 Domain coverage

Table 9 reports the number of nonmissing country-year observations for each of the six domains, along with the pairwise sample sizes for all 15 domain pairs. The six-domain panel spans 17,417 unique country-year rows.

Of the 15 possible pairwise correlations, only six domain pairs have at least 20 overlapping country-year observations and are therefore estimable. The six eligible pairs fall into two clusters: the three large World Bank macro domains (mortality, agriculture, and GDP growth) overlap extensively with each other, while the three narrower domains (natural disasters, climate anomalies, and sovereign credit) overlap among themselves but not with the macro panel. The six pairwise correlations and their sample sizes are reported in Figure 10 of the main paper. Specific pairwise sample sizes for all 15 pairs are available in the replication archive (`crossdomain_six_domain_panel.csv`).

Table 9: Domain coverage and pairwise sample sizes

Panel A: Nonmissing country-year observations by domain

Domain	Nonmissing observations
Agriculture (cereal yield)	9,643
Mortality (death rate)	11,915
Disaster (event count)	3,660
Climate (temperature anomaly)	5,064
Macro (GDP growth)	10,895
Credit ($\hat{\kappa}$)	220
Panel total unique country-years	17,417

Panel B: Pairwise overlapping observations

Domain pair	Pairwise n	Eligible ($n \geq 20$)
Mortality $\times$ Macro	10,895	Yes
Mortality $\times$ Agriculture	9,643	Yes
Macro $\times$ Agriculture	9,390	Yes
Climate $\times$ Disaster	3,239	Yes
Credit $\times$ Disaster	177	Yes
Credit $\times$ Climate	160	Yes
Mortality $\times$ Climate	0	No
Mortality $\times$ Disaster	0	No
Mortality $\times$ Credit	0	No
Macro $\times$ Climate	0	No
Macro $\times$ Disaster	0	No
Macro $\times$ Credit	0	No
Agriculture $\times$ Climate	0	No
Agriculture $\times$ Disaster	0	No
Agriculture $\times$ Credit	0	No

E Benchmark-free estimator diagnostics

E.1 Recovery convention sensitivity

The benchmark-free estimator in Equation (7) of the main paper uses a recovery assumption of 10% for the price-based calculation, matching the convention in the corporate-bond literature. Table 10 reports the rank correlation for alternative recovery values.

Table 10: Spearman rank correlation between benchmark-free and spread-based ordering under alternative recovery assumptions

Recovery rate (δ)	Spearman ρ	95% CI
0.05	0.81	[0.78, 0.84]
0.10 (baseline)	0.82	[0.79, 0.85]
0.20	0.82	[0.79, 0.85]
0.30	0.81	[0.78, 0.84]

The rank correlation is essentially invariant to the recovery assumption: the ordering preservation is a property of the cross-sectional κ distribution, not an artifact of the recovery convention.

E.2 Subgroup consistency

The main paper reports that the benchmark-free ordering is consistent across sukuk ($\rho = 0.82$) and conventional bonds ($\rho = 0.81$). The subgroup estimates use the subset of bonds with an unambiguous sukuk or conventional classification at the snapshot date used for the subgroup analysis; the full canonical estimator contains 856 bonds.

Table 11: Benchmark-free ordering by instrument type

Subgroup	N	Spearman ρ	Median κ (%/yr)
Sukuk	273	0.82	3.12
Conventional	595	0.81	3.78
Unclassified	88	—	—

The 88 unclassified bonds are those whose snapshot classification was ambiguous or unavailable at the analysis date. The classification snapshot (868 bonds with unambiguous classification, 956 total identified) draws on a different vintage than the canonical 856-bond analysis sample; the temporal gap in snapshots accounts for the difference in totals, and all 856 canonical bonds are included in the 956 identified in the classification snapshot.

F Protocol and simulation evidence

F.1 Manipulation resistance

Paper 13’s defense of the published κ against tampering has two parts; the oracle depth rule needs live order-book depth, but its cross-sectional CUSUM monitor is testable on our panel. Reproducing it on 66 issuers (versus the paper’s 37) — fitting each issuer’s κ to the cross-sectional median, then running a Page CUSUM on the level residuals — an injected warp is detected with a median lag of 11 months at 10 bp, 8 at 25 bp and 5.5 at 50 bp (the paper reports 11.5, 9.5 and 5.0), catching 92% of issuers. Targeted κ -warps are caught in months, bigger ones faster — the integrity mechanism holds on previously unused external data.

F.2 Depth rule and manipulation frontier

Paper 13’s other defense — the depth rule, capping repriced notional so a trade cannot move the mark past the validity band — needs order-book depth, which we pulled directly (dYdX and Binance, BTC and ETH). On the deep venues a trade sized at 7% of one-sided book depth moves the mark only 3–13 bp, inside the ± 75 bp clamp, and the dollar cost to push the mark to the 25 bp detection threshold is \$1.3M on dYdX rising to \$26M on Binance. A full Monte-Carlo of the frequent-batch-auction mark against a continuous last-trade mark, on books calibrated to that live snapshot, confirms the FBA mark is orders of magnitude costlier to warp (a multiple of $\sim 1,700\times$ at 10 bp rising past $10,000\times$ at 200 bp, 95% CIs excluding 1), reproducing the prior working-paper sequence’s $72\text{--}2,739\times$ claim in direction and order. Layering the 30-block TWAP, the clamp and the CUSUM monitor on top, a sustained “grind” warp is detected within 0–12 blocks and its net extraction is zero or negative once the cost to hold the position is counted: manipulation does not pay. The measurement-integrity stack, previously argued, is now validated on both live order-book data and simulation.

F.3 Liveness trigger

The validator-liveness sleeve (Paper 12) arms only on a *conjunction* — low transaction volume *and* macro stress — precisely so it cannot be triggered by an ordinary downturn. Pulling dYdX’s TVL history (a \$390M $\rightarrow$ \$96M drawdown, a genuine crypto volume winter) and crossing it with our credit κ -stress index, the two legs never coincide: the volume winter is 2025–26, the κ stress is the 2020 pandemic spike, and the conjunction fires in zero months. The trigger correctly stays shut during a crypto-only winter — the false-positive avoidance the design intends, confirmed on data.

F.4 System-level simulations

The two purely simulated claims — protocol solvency and censorship-resistance — reproduce when their original code is re-run. Executing Paper 3’s cadCAD market (version 0.5.3) over thirty Monte-Carlo runs of 720 steps each gives zero insolvency events in 30/30 runs, an insurance fund that grows from its \$100k seed to \$100.9k (self-sustaining), and a funding rate that never leaves the ± 75 bp clamp — the solvency claim exactly. Re-training Paper 12’s censor-cartel with 150,000 PPO steps, the learned censorship policy earns -0.39 less than honest behaviour and beats honesty in 0 of 30 matched episodes, with suppression converging to 1.4%: censorship is strictly dominated and the liveness sleeve is censorship-proof, as claimed. These are reproductions rather than out-of-sample tests — they re-run the authors’ own simulators — but together with the data-grounded manipulation and conjunction results above they close the last open claims with live code.

F.5 Monetary extensions

The monetary half of the prior working-paper sequence — the κ -Standard (Paper 8) and the interest-free DSGE (Paper 9) — ships replication code but, unlike the pricing papers, had received no independent out-of-sample test. Its quantitative claims are nonetheless measurable, and we test four.

The determinacy ceiling, on twice the sovereigns. Paper 9’s price-level determinacy condition under active-fiscal closure is $\gamma_s < \gamma_s^* = e^{\bar{\kappa}} - 1$, where $\bar{\kappa}$ is the steady-state credit intensity. Reproducing its `empirical_determinacy_map.py` returns the published twelve-sovereign ceilings exactly (Qatar 0.0072 $\rightarrow$ Pakistan 0.0955, ordered by credit quality). We extend it to *twenty-six* distinct sovereigns across two independent instruments: the twenty Cbonds sovereign USD T-spread indices and the twelve sovereign CDS curves (3-year). On both legs the ceiling is small but strictly positive and monotone in credit quality (CDS leg 0.006 Korea $\rightarrow$ 0.139 Egypt, bracketing the paper’s range; the index leg runs 1.3–2.6 $\times$ wider for the same sovereign, the systematic wedge of an index spread over a 3-year CDS). The determinacy window is narrow but never empty for any performing sovereign — the first author-run out-of-sample support of Paper 9’s central result, on an instrument the paper did not use.

Sovereigns and corporates. Extending the map to corporates (Paper 9’s `sovereign_vs_corporate`) on our 107-issuer panel (versus the original 34), the corporate determinacy ceiling sits above the sovereign in five of seven countries with both, and corporate credit is *more persistent* (AR(1) $\rho = 0.865$ versus the sovereign 0.77, half-life 4.8 months), so the buffer requirement is larger:

$$S^* \propto \sigma^{2.3}(1 - \rho)^{-2.2}, \quad S_{\text{corp}}^* \sim 25\%, \quad S_{\text{sov}}^* \sim 12\%.$$

The sovereign–corporate ordering therefore survives on a sample with roughly three times the number of issuers.

Money-base auto-contraction. The κ -Standard prices base money as $M = \sum_i N_i e^{-\kappa_i(1-\delta)T_i}$, which contracts when intensities rise. Reproducing Paper 8’s `full_universe_base.py` gives -1.99% per $+100$ bp on its 195-bond panel; on our 856-bond panel the sukuk sub-panel contracts -2.03% (an essentially exact match) and the full panel -3.08% , the elasticity scaling as $\approx -(1 - \delta)\bar{T}$ with the mean maturity $\bar{T}$ — the countercyclical base mechanism, supported on the expanded sample. The base is moreover sensitive to the curve’s *shape*, not only its level: a steepening (slope) shock moves the base less than a parallel one (-2.3% vs -3.1% per 100 bp) but concentrates the impact in long maturities (the ≥ 10 -year bucket falls 11.8% while short bonds rise), so the κ -Standard carries a term-structure exposure that a single $\bar{\kappa}$ would miss.

Half-lives, systematic share, and the buffer, as ladders. Two reconciliations emerge, each the monetary analogue of the κ -CDS ladder of Appendix C.1. First, κ ’s mean-reversion half-life is construction-specific: per-issuer constant-maturity 2.7 months (Paper 9), per-sovereign index 6.1 months, cross-sectional median index 3.0 months, aggregate index 7.1–8.4 months (Paper 2a) — to be quoted with its aggregation level, not as a single number. Second, the systematic share of κ reconciles cleanly: a principal-components decomposition puts the first component at 63% on all twenty sovereigns, rising to 72% on the seventeen performing sovereigns and 75% on the ten investment-grade core — so Paper 9’s “80–100% systematic” (an index/core statistic) and our 63% (a broad cross-section) are the same quantity at different breadth. The buffer estimate inherits this

sensitivity: because S^* depends on $(1 - \rho)^{-2.2}$, it ranges from $\sim 12\%$ (per-issuer persistence) to $\sim 45\%$ (index persistence), so the prior working-paper sequence’s “11–13%” figure is itself method-specific and should be reported with the persistence measure it assumes.

The real-block feedback exists. Paper 9’s determinacy also turns on the procyclical credit channel $\hat{\kappa} = \psi(-\hat{y})$, whose threshold it places near $\psi^* \approx 0.62$. Pooling 159 country-years of World Bank GDP growth against the next change in credit- κ , the feedback is strongly procyclical and significant ($\rho = -0.50$, $p < 0.001$; κ rises as output falls). Its raw magnitude is unit-sensitive (the paper’s $\hat{y}$ is an output gap, not a growth rate), so we read the result as confirming the *sign and strength* of the channel rather than a precise ψ — and the strength is itself the point: the feedback is large enough that the paper’s stabilisation devices (net-worth predetermination, Tobin’s Q, a sovereign stabilisation fund), which it shows restore determinacy, are empirically relevant rather than merely formal.

G Analysis hierarchy, exploratory tests, and null results

G.1 Analysis hierarchy

The primary validation ledger contains 81 analyses across the main paper and appendices. The analysis hierarchy was recorded in a dated internal plan dated March 2026, which classified each analysis as primary (confirmatory), secondary (descriptive), or exploratory. The plan was not externally preregistered; exploratory analyses are labelled separately in the replication archive. Appendix G.2 separately summarises 30 additional exploratory extensions outside the 81-analysis ledger.

Table 12 summarises the distribution of analyses by category and outcome.

Table 12: Summary of the 81-analysis ledger

Category	Confirmed	Null/refuted	Total
Primary (pre-specified)	16	2	18
Secondary (descriptive)	31	4	35
Exploratory (planned, not pre-specified)	22	6	28
Total	69	12	81

G.2 Exploratory analyses beyond the pre-specified set

Following the 81-analysis ledger, we ran a further set (catalogued A91–A120 in the replication archive) extending the validation in three directions — the multi-modal oracle, the interest-versus- κ system comparison, and the dynamics and economics of κ . We summarise the findings here at the level of conclusions; the scripts and full output are in the archive. We report the null and refuted results alongside the confirmations, because a validation exercise that omits failures would not be informative.

κ as a hazard — supported in the reported sample.

- *Convergence is universal:* the basis mean-reverts in 43/43 perpetual markets, median half-life 3.7 h, reproducing the BTC keystone measurement market by market (A114).
- *Out-of-sample default prediction:* the three sovereigns that actually defaulted (Lebanon, Sri Lanka, Ukraine) sat in the top decile of the cross-sectional κ distribution twelve months before default — 3/3 at every horizon (A113).
- *Mean-reverting intensity, not a random walk:* forward κ regresses on its own deviation with slope -0.26 , AR(1) half-life ~ 3.4 months (A109).
- *Cross-instrument and cross-asset coherence:* CDS- κ and bond- κ for the same sovereign co-move at 0.69 in changes (A111); sovereign- κ co-moves with the currency at +0.34 (A112).
- *Within-rating information:* inside a rating bucket, κ deviation predicts the forward spread change at corr -0.47 — κ differentiates risk the static letter cannot (A119).
- *The risk premium is earned:* realised one-year excess returns rise monotonically with κ (low +0.6%, high +7.4%), evidence that bearing κ -risk is compensated — gain accompanying liability for loss (A116).

- $\sim 67\%$ of sovereign- κ variation is one systematic factor (A120).

Interest is removable, with measured consequence — supported in the reported sample.

- The interest parameter is a committee choice (the policy rate is unchanged 47% of months, then jumps by decree) while κ is a continuous measurement; their correlation is -0.76 (A99).
- The practical payoff is the 2022 duration gradient: conventional bonds lost up to -45% at thirty years to a rate decision, while the $\iota = 0$ formula carries no direct rate term at any tenor (A100; a model property, with the mixed-economy equilibrium caveat stated in the main paper).
- Across the panel rate risk is 40–52% of realised yield volatility, ballooning in the rate-shock year (A108).
- At $\iota = 0$ the funding distribution has no formula-induced mass point around one basis point, consistent with the premium-only design (the dYdX nine-market equal-weight signed mean is 0.83 bp per complete eight-hour interval); the distribution is continuous rather than pinned at a fixed-input value (A110).

Multi-modal oracle security — supported in the reported sample.

- The three recovery channels are nearly mutually independent ($\rho \approx 0.05$, A91); a crisis moves only the channels in its causal path (2020 lit up credit and mortality but not crops; 2018 the reverse, A92–A95); and a median of three channels with two non-tradeable legs converts a $\sim \$23\text{M}$ market attack into an effectively-infeasible two-institution data breach (A96).

Null and refuted results — reported honestly.

- The macro claim that interest-debt drags growth is *not* supported once development confounds and reverse causality are removed (within-country, lagged: the relationship collapses to zero, A107).
- κ does *not* lead the currency — FX moves first (A112).
- κ 's monthly changes show *no* clean up-jump asymmetry (the extreme skew is an outlier artifact, A117).
- Sukuk and conventional κ are highly correlated (0.91) but *not* identical — a ~ 1 pp level gap from maturity and demand, most of which is maturity (A118).
- The cross-country κ –inflation correlation ($+0.61$) is a fragility confound, not κ containing inflation (A103).

These results delimit the supported claims and record the failures alongside them.

G.3 Cross-domain null results

The cross-domain analysis in the main paper and Appendix D reports a mean absolute off-diagonal correlation of 0.075 among six estimable pairs. The null $\rho_x = 0$ is not rejected for any individual pair at conventional significance levels after Bonferroni correction for the planned family of 15 pairs. The small magnitudes are therefore consistent with approximate independence among the six estimable domain pairs in the canonical construction — the assumption that underlies the κ -ladder diversification argument.